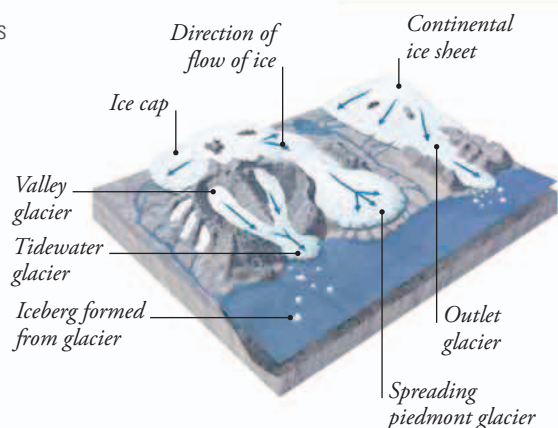


Glaciers and ice shelves

In cold climates, snow stays frozen throughout the year, so it gets deeper and deeper as more snow falls. Its weight compresses the lower layers of snow into solid ice, which flows slowly downhill as glaciers. Many of these melt before they reach the coast, but some polar glaciers flow all the way to the sea. Here they form the tidewater glaciers and ice shelves that break up to create icebergs.

TYPES OF GLACIERS

Most glaciers form in the mountains from snow that collects in a rock basin between high peaks. When the basin is full, the ice overflows and grinds its way downhill as a valley glacier. But in very cold regions, high land is covered with ice caps or huge ice sheets that feed ice into outlet glaciers. Both types can reach the sea, where they become tidewater glaciers.



Hubbard Glacier
Crumbling cliffs

Location Alaska
Length 76 miles (122 km)
Status Advancing

This is the biggest tidewater glacier in North America. It has a huge tidewater front that extends for 6 miles (10 km), with an ice cliff up to 394 ft (120 m) high. Ice crumbling from the cliff forms a steady stream of icebergs that drift into Disenchantment Bay on the southeast Alaskan coast. Despite this, the glacier is regularly visited by cruise ships.



Jakobshavn Glacier
Giant icebergs

Location Greenland
Length More than 40 miles (65 km)
Status Retreating

The Jakobshavn Glacier is one of many outlet glaciers pouring off the Greenland ice sheet—a mass of ice that covers 80 percent of Greenland to a depth of up to 1.9 miles (3 km). The glacier flows west into a fjord that opens into Disko Bay, breaking into huge icebergs that then drift into the north Atlantic. The iceberg that sank the *Titanic* in 1912 probably came from this glacier.

Columbia Glacier
Iceberg factory

Location Alaska
Length 32 miles (51 km)
Status Retreating

One of the many tidewater glaciers that spill into the Gulf of Alaska in the north Pacific, this is one of the fastest-moving glaciers in the world. In 2001, it was calving icebergs into the sea at the rate of 1.7 cubic miles (7 cubic km) per year. But this prolific iceberg production is cutting back the floating ice front, which has retreated to 10 miles (16 km) since 1982.

